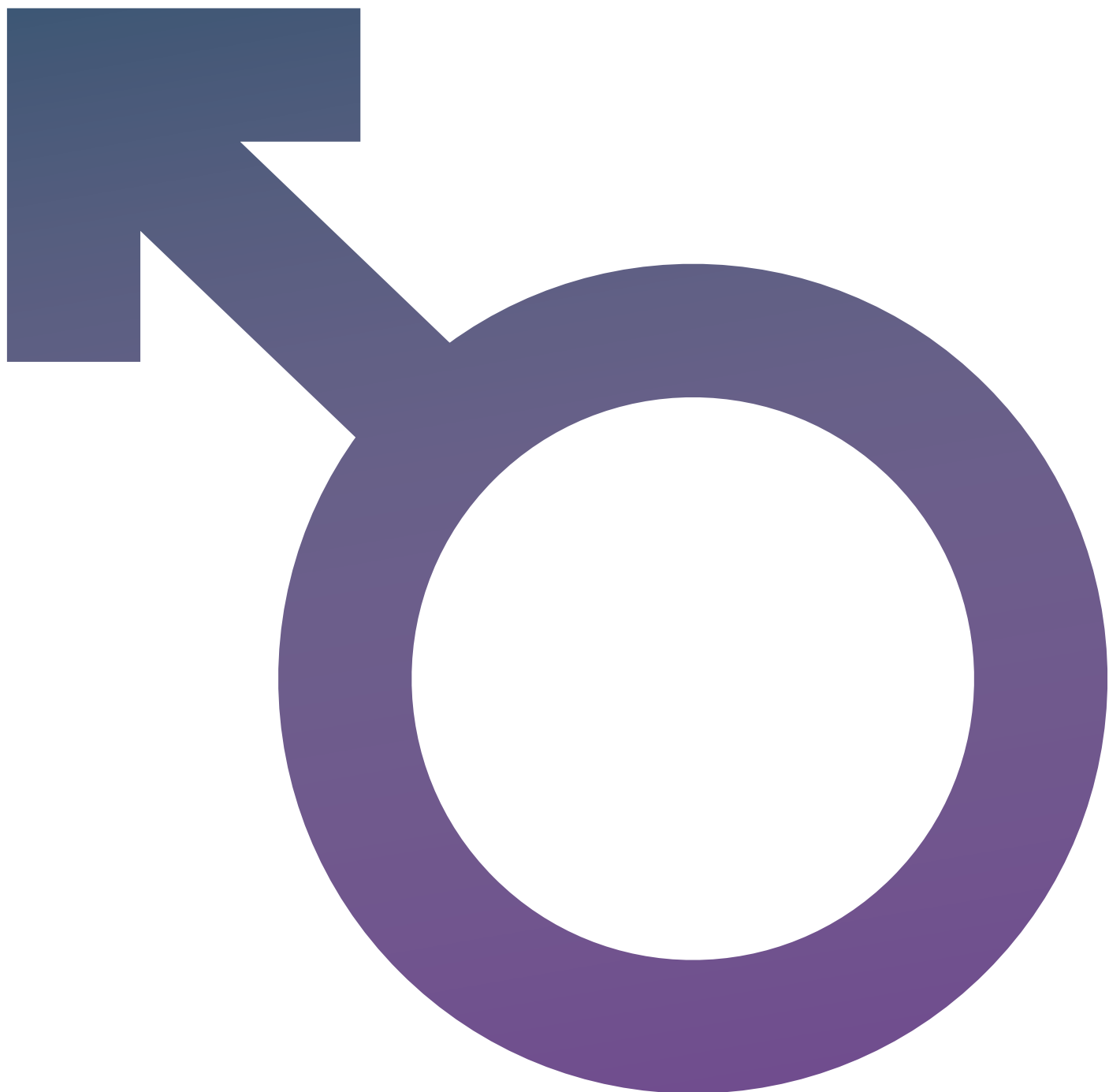


Examine.com

Testosterone Stack Guide



Medical Disclaimer

This guide is a general health-related information product intended for adults over the age of 18.

This guide is for educational purposes only. It does not constitute medical advice. Please consult a medical or health professional before you begin any exercise, nutrition, or supplementation program, or if you have questions about your health.

If you choose to engage in any activity or use any product mentioned in this guide, you do so of your own free will, and you knowingly and voluntarily accept the risks.

While we mention major known interactions, it is possible for any supplement to interact with other supplements, as well as with foods and pharmaceuticals. Therefore, it is important to consult a medical professional prior to using any supplement mentioned in this guide.

Specific study results mentioned in this guide should not be considered representative of typical results. Individual results do vary.

Keep in mind that not all supplements contain the exact compounds and amounts listed on the label. Always investigate supplement companies, as well as the supplement itself, before purchasing anything. Also note that, more than isolated compounds, herbs can have variability from batch to batch, which can alter their efficacy and safety.

For evidence supporting the claims mentioned in this guide, please visit [Examine.com](https://www.examine.com).

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How to Use This Guide

The team at Examine.com has been publishing research on nutrition and supplementation since March 2011. Drawing from all we've learned, we designed this Stack Guide to help you figure out which supplements can help you reach your health goal, and which can hinder you or just waste your money.

Core supplements are the supplements most likely to help, while having little to no side effects. They tend to have more research backing than do the other supplements.

Primary options may provide substantial benefit, but only in the right context. A primary option is not for everyone, but if you read the entry and find that you meet the criteria, consider adding the supplement to your stack.

Secondary options form another group of potentially beneficial supplements, but with less evidence for their effects. They could work or be a waste of money. Keep them in mind, but think twice before incorporating them into your stack.

Inadvisable supplements have either been shown to be ineffective, marketing claims notwithstanding, or are considered too risky. Do not add them to your stack. At best, they will be a waste of money; at worst, they can cause you harm.

Now that you have been presented with various supplements worthy of your interest, the time has come to combine them based on your objective. We will guide you in **assembling your stack**.

Then comes the **FAQ**, in which we cover common questions that may arise when assembling your stack.

Lastly, we include information on **precautions and troubleshooting**.

With all this combined, you should be able to identify and assemble the supplement stack best suited to your objective.

Core Supplements

Magnesium

Why it's a core supplement

Supplementing magnesium (Mg) can help normalize testosterone levels, but only when low levels are linked to a magnesium deficiency. If your body has enough magnesium, taking more will not benefit you.

How to take it

A diet comprising magnesium-rich foods (such as fish, nuts, beans, and green leafy vegetables) renders supplementation unnecessary, at least for the purpose of testosterone normalization. In case of magnesium deficiency, adding or increasing [dietary sources of magnesium](#) should be the first option, but in the absence of practical ways of doing so, supplementation can be used.

Commonly supplemented forms of magnesium include magnesium gluconate, diglycinate, and citrate. To increase absorption, magnesium gluconate should be taken with a meal, while other forms of magnesium can also be taken on an empty stomach. Because of its low bioavailability, magnesium oxide can cause intestinal discomfort and diarrhea; it is therefore not recommended for supplementation.

The standard dose is 200 mg of elemental magnesium once a day, though up to 350 mg can be used. Because magnesium might have a sedative effect, it is often supplemented before bed.

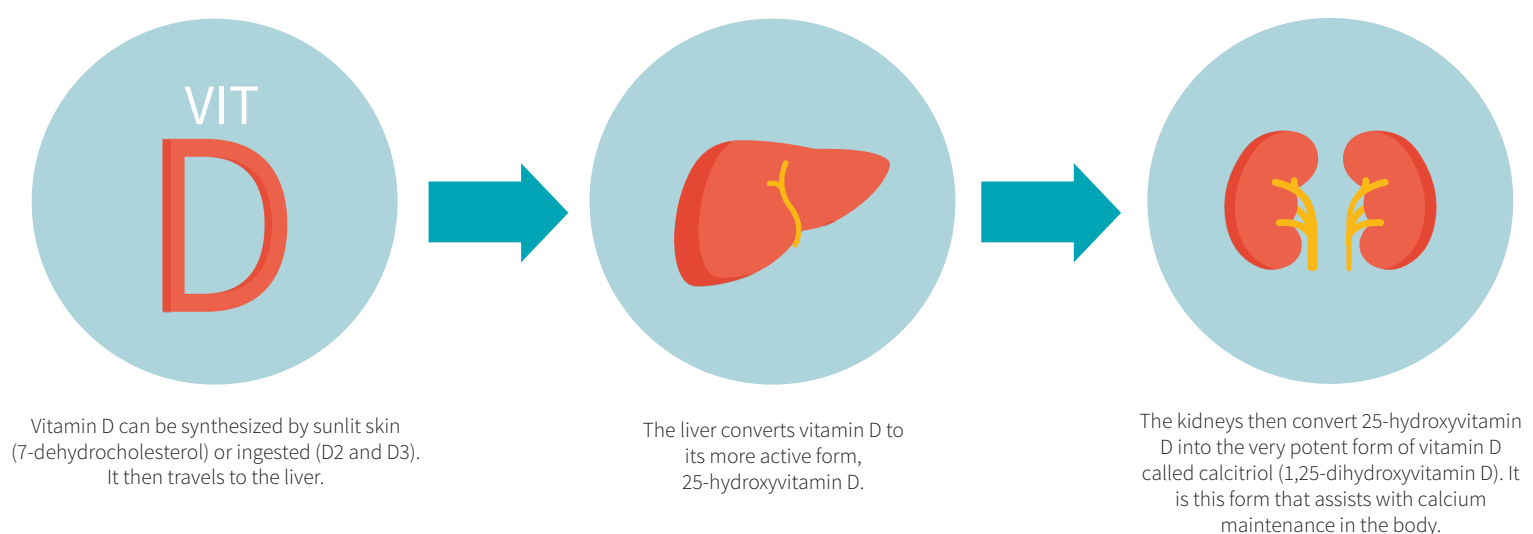
Avoid taking [calcium](#), [iron](#), magnesium, and [zinc](#) at the same time in combinations of 800+ mg, since high amounts of these minerals will compete for absorption. Like zinc, magnesium may also impair the absorption of antibiotics, notably those in the tetracycline class (e.g. doxycycline) and quinolone class (e.g. ciprofloxacin), so take magnesium and antibiotics at least 6 hours apart.

Vitamin D

Why it's a core supplement

Vitamin D deficiency is common, especially in people whose exposure to sunlight (without clothes or sunscreen) is limited. Moreover, the darker your skin, the longer you need to expose yourself to sunlight to synthesize enough vitamin D. Still, if you spend a lot of time outside and live near the equator, you can skip supplementation!

Figure 1: How vitamin D is made



Vitamin D has often been researched in the context of male fertility.

Vitamin-D receptors can be found on sperm cells, and vitamin D may also play a role in the production of steroid hormones. In a study on men with low vitamin-D levels, supplementing vitamin D over the course of a year led to an increase in testosterone levels. Since the participants were all middle-aged men, however, they may have started the experiment from a state of age-related testosterone decline. The effect of vitamin-D supplementation on younger men is currently unknown.

How to take it

Take 2,000–3,000 IU (50–75 mcg) of **vitamin D₃** with with a meal containing fat, either year round or only during the colder, darker months, when you are least likely to synthesize enough vitamin D from sun exposure.

Doses higher than 3,000 IU may be warranted in cases of severe deficiency or non-response at lower doses, as ascertained by a blood test. Keep in mind that 10,000 IU/day for three months has been shown to be toxic.

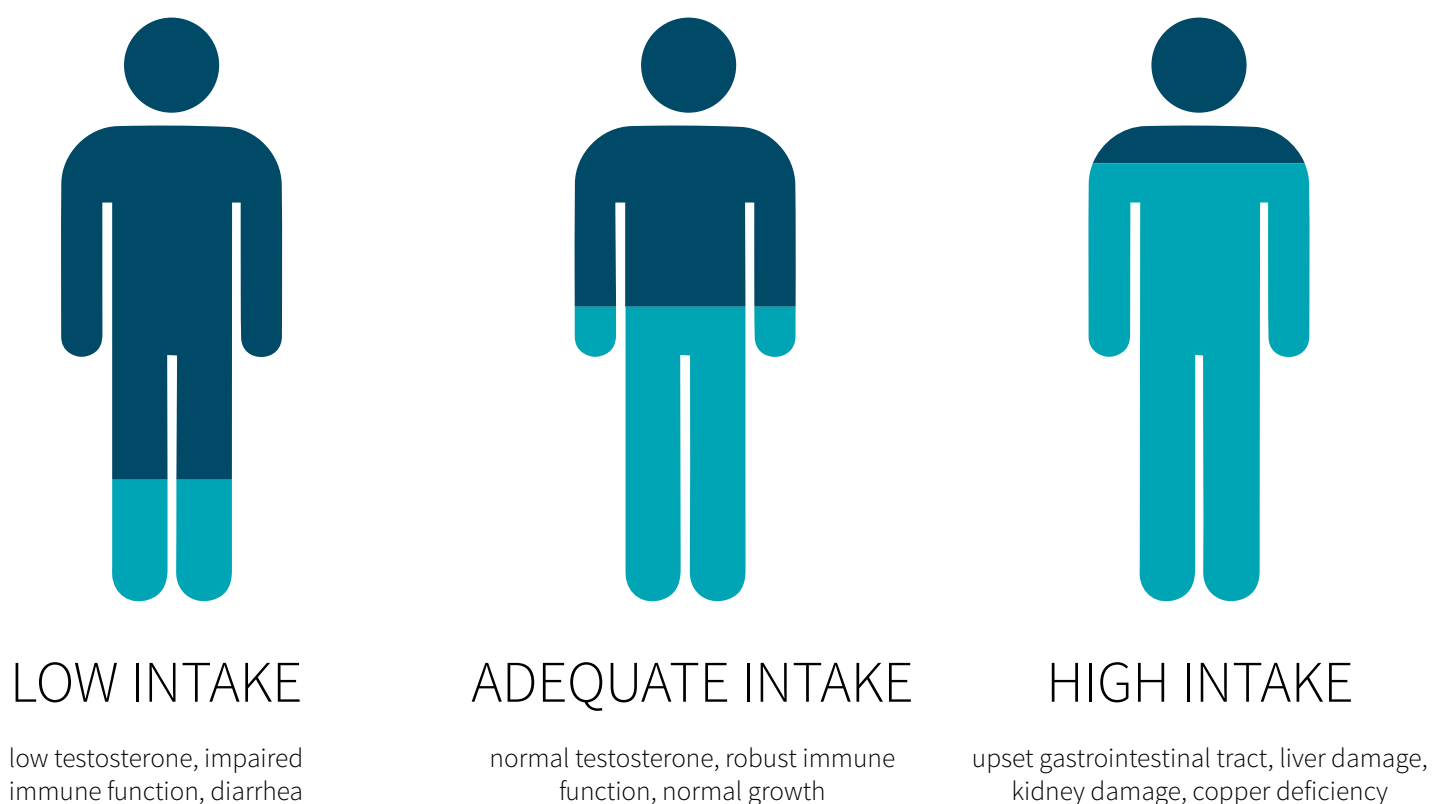
Zinc

Why it's a core supplement

Zinc (Zn) is an important mineral for general health and is often marketed as a testosterone booster. As with [magnesium](#), however, zinc supplementation can only help when low testosterone levels are linked to a zinc deficiency. If your body has enough zinc, taking more will not benefit you.

On the contrary, over time, high doses of zinc can irritate the gastrointestinal tract. They can also cause a copper deficiency, since zinc kick-starts the process of creating metallothionein, a protein that binds zinc but also other metals, notably copper; the bound metals then leave the body as waste products. Even higher doses of zinc can also damage the liver and kidneys, but the doses in this guide are too low to pose any of those risks.

Figure 2: Effects of low, adequate, and high zinc intakes



How to take it

Zinc requirements vary according to diet and level of activity. Sedentary people who do not sweat much and eat enough meat might not need to supplement zinc at all, and should otherwise limit themselves to 10–20 mg/day (15–25 mg/day for vegetarians and vegans). Athletes and other people who sweat a lot (which results in zinc loss) can take 25–30 mg/day.

Zinc should be taken with meals, so as to prevent potential nausea. Avoid taking [calcium](#), [iron](#), [magnesium](#), and zinc at the same time in combinations of 800+ mg, since high amounts of these minerals will compete for absorption. Though to a lesser extent than magnesium, zinc may also impair the absorption of antibiotics, notably those in the tetracycline class (e.g. doxycycline) and quinolone class (e.g. ciprofloxacin), so consider taking zinc and antibiotics at least 6 hours apart.

Primary Options

DHEA

Why it's a primary option

The hormone dehydroepiandrosterone (DHEA) circulates throughout the body and can be called to make other hormones, notably testosterone and estrogens. Supplemental DHEA can support normal testosterone levels; this effect is especially reliable in case of age-related low testosterone. People with healthy hormone levels will see no benefit from supplemental DHEA.

As DHEA is a mild CYP3A4 inhibitor, it should not be used alongside pharmaceuticals that are metabolized by this enzyme.

How to take it

Take 25–50 mg of DHEA once a day with a meal.

Secondary Options

Boron

Why it's a secondary option

Like [magnesium](#) and [zinc](#), boron is a dietary mineral. As a supplement, it is sometimes recommended for postmenopausal women with low hormone levels. Some studies have also noted an increase in testosterone in men, even in young men, but this effect is unreliable — its magnitude varies significantly between studies using similar dosages and methodologies. More research is needed to ascertain boron's effects on hormone levels and determine the mechanism behind those effects.

How to take it

Studies have noted an increase in the testosterone levels of young men taking 10 mg of boron per day. Do not take more than 20 mg/day.

Coleus Forskohlii

Why it's a secondary option

Coleus forskohlii is an herb historically used in Ayurvedic medicine. It can increase the body's levels of cyclic adenosine monophosphate (cAMP). It might also increase the efficacy of other supplements that increase cAMP, such as [caffeine](#). An increase in cAMP levels in the testicles leads to an increase in testosterone synthesis. Supplementation of *Coleus forskohlii* has been noted to increase testosterone synthesis in one study on overweight men.

Because of its many bioactive compounds and a lack of information on its long-term effects, *Coleus forskohlii* should be cycled. It should not be taken with blood thinners, calcium-channel blockers, or pharmaceutical nitrates (supplementing nitrates through foods, such as beetroot juice, is fine), as this could result in hypotension (low blood pressure). In addition, it might not be safe for people suffering from liver ailments, as a rodent study noted an increase in liver fat (not due to forskolin but to other, unknown compounds in the herb).

Because of those caveats, and the limited number of randomized controlled trials in humans, *Coleus forskohlii* is only a secondary option.

How to take it

The main bioactive compound in *Coleus forskohlii* is called forskolin. More research is needed to confirm an ideal dosage, but the standard protocol is 25 mg of forskolin twice a day, 4 to 6 hours apart, for a total of 50 mg/day. Most *Coleus forskohlii* extracts are 10% forskolin, in which case 25 mg of forskolin translates to 250 mg of the extract.

Coleus forskohlii supplementation should last no longer than 12 weeks, followed by a 12-week break. Short-term use has been shown to be safe, but more research is needed to determine the long-term effects.

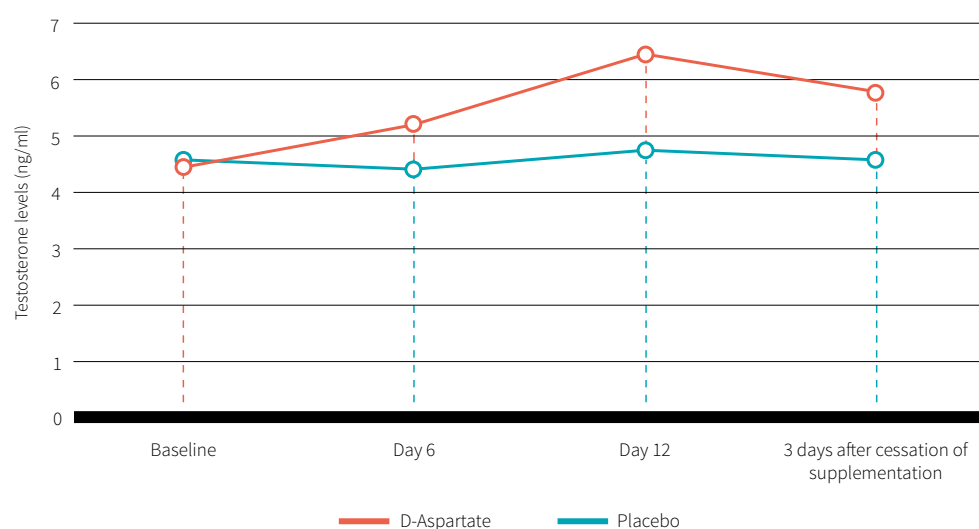
D-Aspartic Acid

Why it's a secondary option

D-Aspartic acid (DAA, or D-aspartate) increased testosterone levels in two studies. The one in healthy men reported a 42% increase with 3.12 g/day. The one in infertile men, which suffered from severe methodological limitations, reported a 30–60% increase with 2.66 g/day. However, two later studies in resistance-trained men noted no increase with 3 g/day; the latest even noted a 12.5% decrease with 6 g/day.

Research is still too scarce and results too inconsistent for DAA to be more than a secondary option. What little evidence there is seems to indicate that men with low testosterone levels are more likely to benefit from DAA supplementation.

Figure 3: Effect of D-aspartate on testosterone release in healthy males



How to take it

Take D-aspartic acid (3 g/day) on an empty stomach or with a meal low in protein, in the morning or before exercise. Supplement for 2 weeks, then take a 6-week break.

Inadvisable Supplements

Creatine

Most studies indicate that creatine has no effect, positive or negative, on testosterone levels. One study did report a minor, transient increase in testosterone post-workout; this spike was probably due, not to creatine directly, but to the increase in training volume made possible by creatine.

Libido Enhancers

People often assume that an increase in libido reflects an increase in testosterone. This is not always true. [Maca](#), [fenugreek](#), and [Tribulus terrestris](#) are among the herbs that can affect your libido yet have no effect on your testosterone levels.

Some other herbs, such as [ginger](#) and [Eurycoma longifolia](#), can only support the testosterone levels of infertile men (or of men with testicular damage).

Many others herbs, like [horny goat weed](#), have not even been studied in humans yet.

Stinging Nettle

Studies have found that stinging nettle can benefit men suffering from benign prostatic hyperplasia, with no effect on their total testosterone levels. Their free testosterone levels, however, were not assessed.

In our bodies, testosterone is either bound (strongly to SHBG, weakly to albumin) or free. SHBG-bound testosterone cannot affect us. A lignan called 3,4-divanillyltetrahydrofuran, found in stinging nettle roots, binds to SHBG *in vitro* (i.e. in test-tube studies). Should this effect extend to oral supplementation, an increase in free testosterone would logically ensue, but

human studies are needed to validate this theory. Moreover, this increase would probably be temporary; with time, the body can be expected to either decrease its production of testosterone or, more likely, increase its production of SHBG (sex-hormone-binding globulin).

Assembling Your Stack

Incorporating Core Supplements

With a meal containing fat, take [zinc](#) (10–30 mg), [magnesium](#) (200–350 mg), and [vitamin D₃](#) (50–75 mcg / 2,000–3,000 IU).

If you get plenty of sun, you may not need to supplement vitamin D₃. Similarly, if your diet is rich in zinc or magnesium, you may not need to supplement these minerals. If you want to supplement both zinc and magnesium, you could do it through [ZMA](#), a supplement that combines magnesium, zinc, and vitamin B₆. Vitamin B₆ does not interfere with the effects of magnesium and zinc, and it may help prevent mineral losses to sweat.

The core supplements are recommended for most people; their efficacy and safety are backed by a significant body of evidence. Take them for a couple of weeks before you consider making any modification, such as adding one of the following options.

Incorporating Options

For young men (<35 years old) who want to increase their testosterone levels

With a meal containing fat, take [zinc](#) (10–30 mg), [magnesium](#) (200–350 mg), and [vitamin D₃](#) (50–75 mcg / 2,000–3,000 IU).

Take [D-aspartic acid](#) (3 g/day) and [Coleus forskohlii](#) (25 mg of forskolin twice a day) for 2 weeks, then take a 6-week break.

For middle-aged men who want to support their testosterone levels

With a meal containing fat, take [zinc](#) (10–30 mg), [magnesium](#) (200–350 mg), and [vitamin D₃](#) (50–75 mcg / 2,000–3,000 IU).

Supplementation should complement a healthy diet and exercise plan.

Other Options

Only a minority of studies noted an increase in testosterone from [creatine](#) supplementation, and that increase was small. However, since creatine is safe and can benefit other aspects of your health (notably your strength and your methylation status), you could elect to add 5 g twice a day (i.e. 10 g/day) to any testosterone stack.

Though [boron](#) is mentioned in this guide, it is not recommended for any stacks due to a paucity of human evidence for its effects.

FAQ

Can I add to my stack a supplement not covered in this guide?

Supplement your current stack for a few weeks before attempting any change. Talk to your doctor and [research each potential new addition](#) in advance. Check for known negative interactions with other supplements in your current stack, but also for synergies. If two supplements are synergistic or additive in their effects, you might want to use lower doses for each.

Can I modify the recommended doses?

If a supplement has a recommended dosage range, stay within that range. If a supplement has a precise recommended dose, stay within 10% of that dose. Taking more than recommended could be counterproductive or even dangerous.

Should I take my supplements with or without food? In the daytime or the evening?

Certain supplements have strong evidence for how and when to take them, which we have noted where applicable. For most supplements, however, this research is either mixed or absent. It is always a good idea to start with a low dose of a supplement, so as to minimize the harm of taking it during the day (e.g. tiredness) or in the evening (e.g. insomnia).

Keep in mind that taking a supplement with food does not mean that all foods will have the same effect on its absorption, digestion, or metabolism. The most basic example is fat-soluble vitamins (A, D, E, and K), which will absorb better with a small meal containing fat than a large meal containing little to no fat.

What's the difference between elemental magnesium/zinc and other kinds of magnesium/zinc?

“Elemental” refers to the weight of the mineral by itself, separately from the compound bound to it. For instance, consuming 500 mg of magnesium gluconate means consuming 27 mg of elemental magnesium, while consuming 50 mg of zinc gluconate means consuming 7 mg of elemental zinc. **Product labels display the elemental dosage.** On a label, “27 mg of magnesium (as magnesium gluconate)” means 27 mg of elemental magnesium (and 473 mg

of gluconic acid), while “7 mg of zinc (as zinc gluconate)” means 7 mg of elemental zinc (and 43 mg of gluconic acid).

I have an iron stomach. I have never felt nauseous from supplements. Do I still need to take precautions to avoid gastrointestinal upset?

If you have never had any issues with nausea or vomiting, you may have an easier time ingesting large doses of certain supplements. Nevertheless, it is not a good idea to disregard the warnings on a product.

Since the body can use DHEA to make estrogen, shouldn't men avoid supplementing it?

In addition to being a signalling molecule in and of itself, DHEA is the precursor to androgens *and* estrogens. Therefore, yes, DHEA may increase estrogens levels in men, depending on various factors, such as an individual's overall levels of androgens and estrogens. However, feminizing effects haven't been reported.

Will caffeine affect my testosterone levels?

Caffeine increases cyclic adenosine monophosphate (cAMP) levels in humans. An increase in cAMP levels in the testicles leads to an increase in testosterone synthesis. So it was logical to think that caffeine might increase testosterone levels in humans. Alas, when a study tested this theory, what increases it noted were minor and highly variable between participants.

Will my diet affect my testosterone levels?

Yes, and in too many ways to count! To tackle the big issues, though, you should keep in mind that diets very low in calories or fat are likely to reduce your testosterone over time.

Precautions and Troubleshooting

Stack components are seldom studied together. The safest way to add supplements to your daily routine is one at a time, at least a couple of weeks apart, to better assess the effects (and side effects) of each new addition. Start at half the regular dose for a week, then slowly increase to the regular dose if you are not experiencing the desired effects. Remember, however, that even if a supplement can coax your body into producing more testosterone, it can only do so within your physiological limits; so do not expect steroid-like effects.

An increase in testosterone is usually followed by an increase in dihydrotestosterone (DHT), which can worsen hair loss in men predisposed to male-pattern baldness (receding hairline and hair thinning on the crown). Should this be a concern, consider using topical minoxidil. You could also ask your doctor about finasteride — which inhibits 5-alpha reductase, an enzyme responsible for the conversion of testosterone to DHT — but keep in mind the potential side effects of medications that alter hormone levels.